Road Network Description File Annotation

1. Course Content

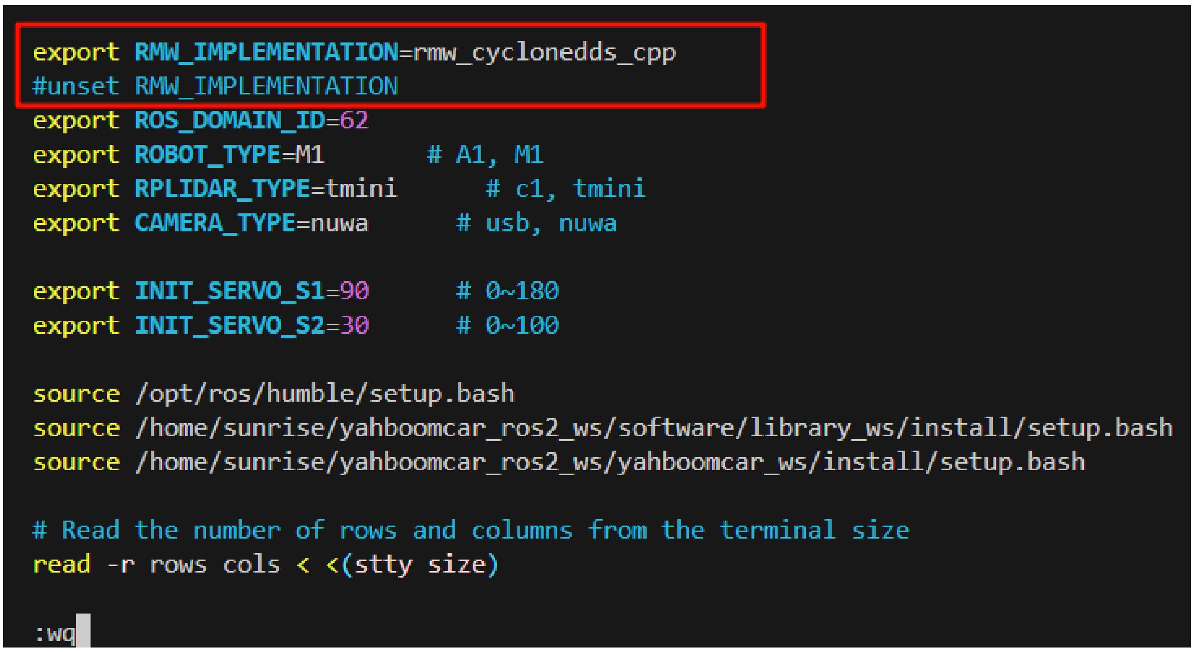
1. Using the route_tool tool to annotate the .geojson format file describing the road network

> [!IMPORTANT]
>
> - rviz visualization is used, all running in the virtual machine `Rosmaster-RoadNet-Ubuntu24.04`

2. Road Network Annotation

Open ~/.bashrc, uncomment this line, and change it as shown in the image. (After learning the road network planning function, please restore this comment to avoid affecting the previous examples.)

```
:wq #Save and exit
source ~/.bashrc #Update environment
```



2.1 Starting the Navigation Node

- When marking the road network, we need to know the exact location of the vehicle's navigation system on the map. Therefore, we need to use the global positioning of the navigation node here.
- (Host side) Chassis radar driver

```
ros2 launch yahboomcar_nav laser_bringup_launch.py
```

- (Virtual machine side) Start visualization

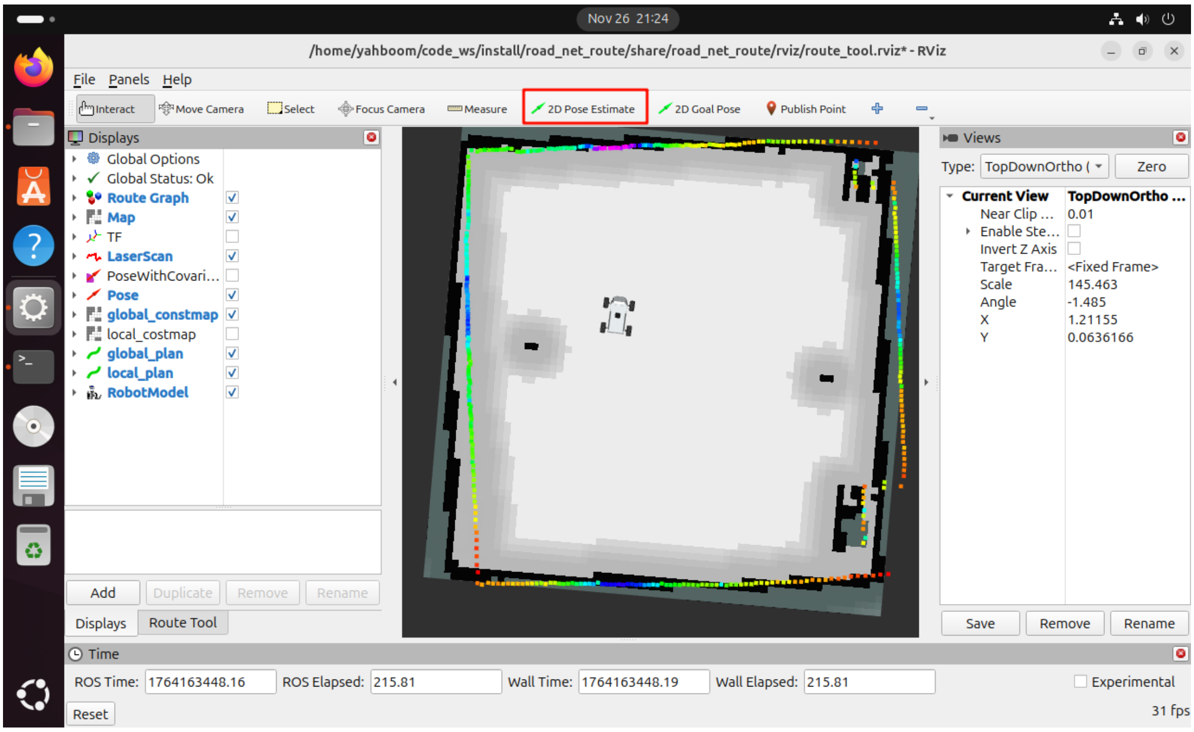
```
ros2 launch road_net_route rviz_only.launch.py ••tools:=true
```

- (Host side) Navigation node

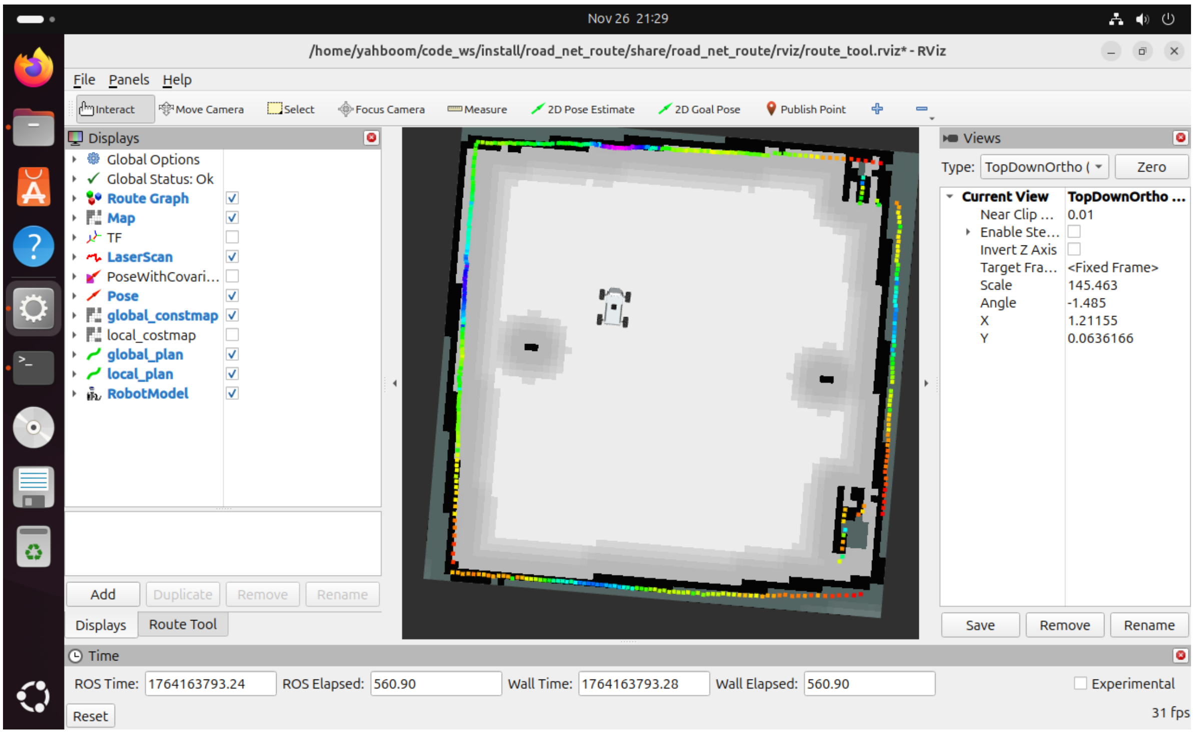
```
#rdkx5
ros2 launch yahboomcar_nav navigation_teb_launch.py
map:=/home/sunrise/map/map.yaml
#orin
ros2 launch yahboomcar_nav navigation_teb_launch.py
map:=/home/jetson/map/map.yaml
```

> [!NOTE]
>
> Startup parameter description
>
> - **map** The name of the raster map to be loaded; here, we've selected the map saved in the previous lesson.
> - The host machine's map is saved by default in the `~/map` path. This path is automatically mounted into the `/root/map` directory within the `ros2_kilted` container. Therefore, map files saved on the host machine can be used directly within the `ros2_kilted` container.

- RViz will display the map and two coordinate systems. `map` is the map coordinate system, and `base_footprint` is the coordinate system of the projection point of the vehicle's chassis center onto the ground. We first use `2D Pose Estimate` to give the vehicle's initial pose on the map we created, roughly its position on the map.

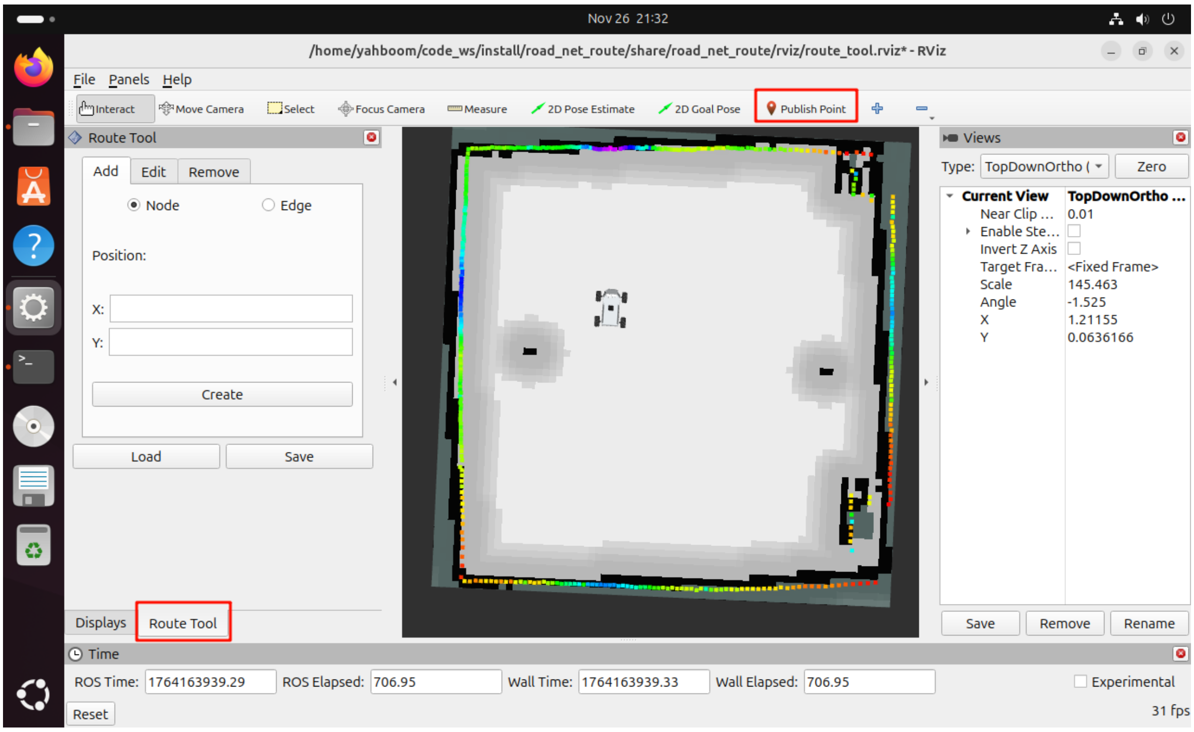


- After giving the approximate position, we can navigate the mobile robot vehicle using keyboard control, gamepad control, or `2D Goal Pose` in RViz, ensuring the radar/laser contour perfectly matches the map.

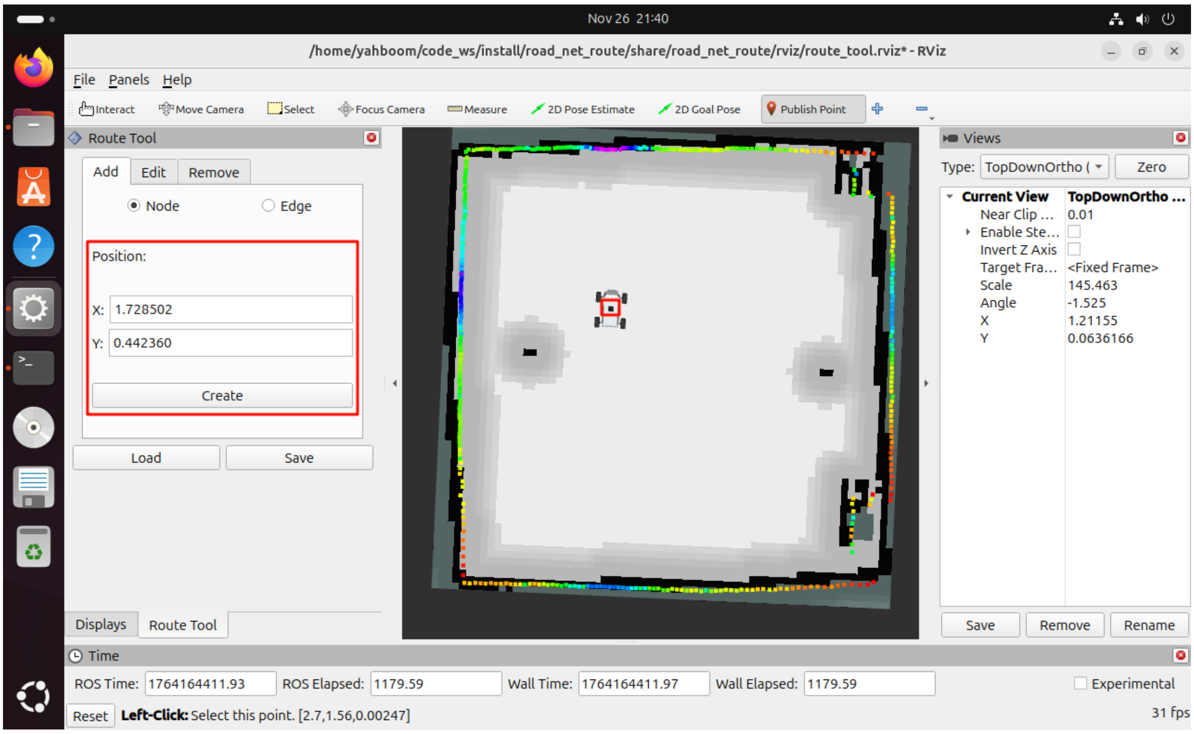


2.2 Adding a Waypoint

- Click the `Publish Point` tool in the RViz toolbar to enter waypoint selection mode.

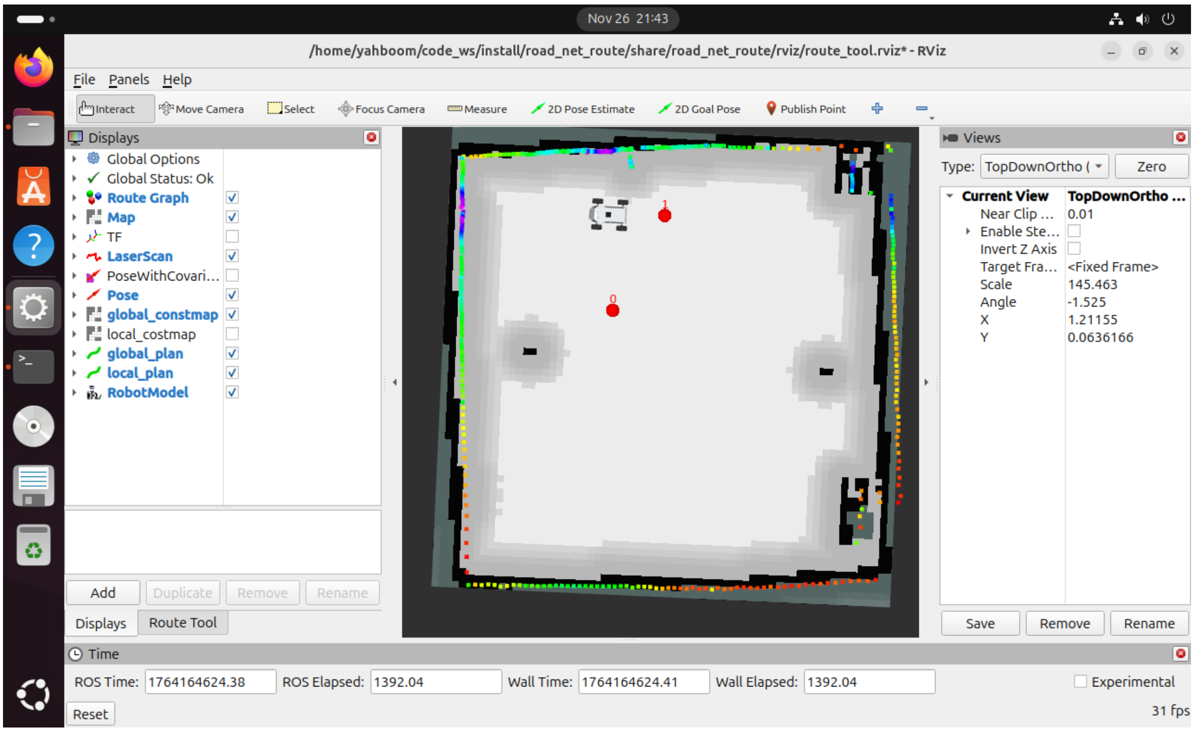


- Then click the left mouse button in the center of the car. The `position` field in the `Route Tool` tab at the lower left will automatically fill in the coordinates of the current point. Then click `Create` to create a waypoint.



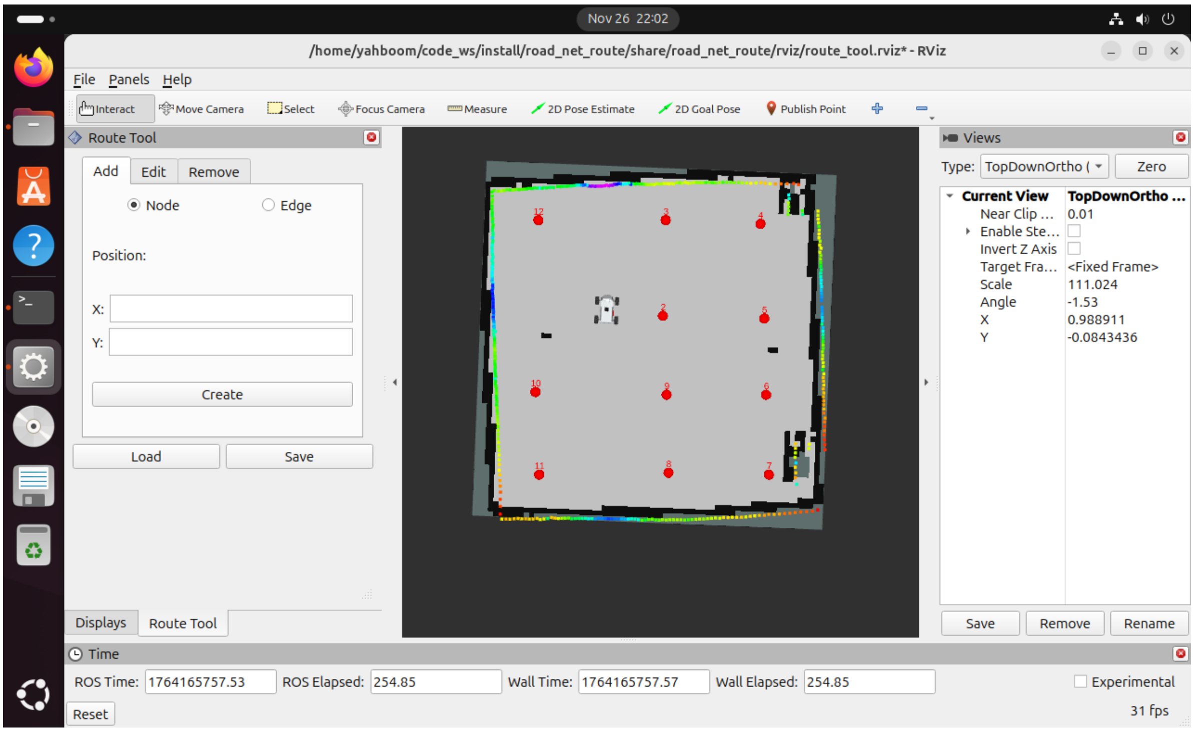
- Next, we move the vehicle-mounted robot to the next location where waypoints need to be marked. In this demonstration, we directly move the robot to the next location where waypoints need to be marked.

> [!TIP]
> The mobile robot-mounted robot can be controlled via keyboard, gamepad, or RViz's `2D Goal Pose` navigation.



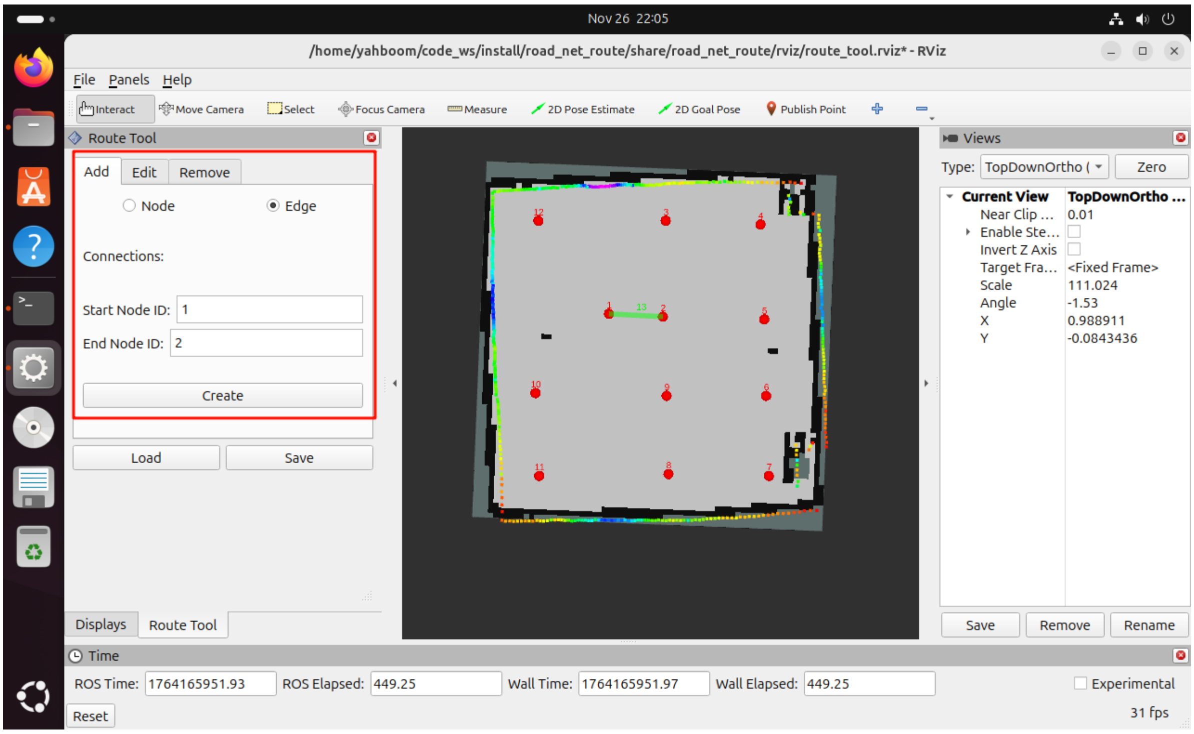
> [!WARNING]
>
> - The localization principle here uses AMCL global localization and pose map matching.
> - **Note:** If, during subsequent waypoint marking, a discrepancy is observed between the laser contour and map environment features (due to cumulative errors in the odometer or wheel slippage causing local positioning errors), you can first move and rotate the robot a short distance to allow AMCL localization to take effect and ensure the radar laser contour data perfectly matches the map.

- Repeat the above steps until all waypoints need to be marked (mark waypoints according to your actual needs).



2.3 Adding Edges

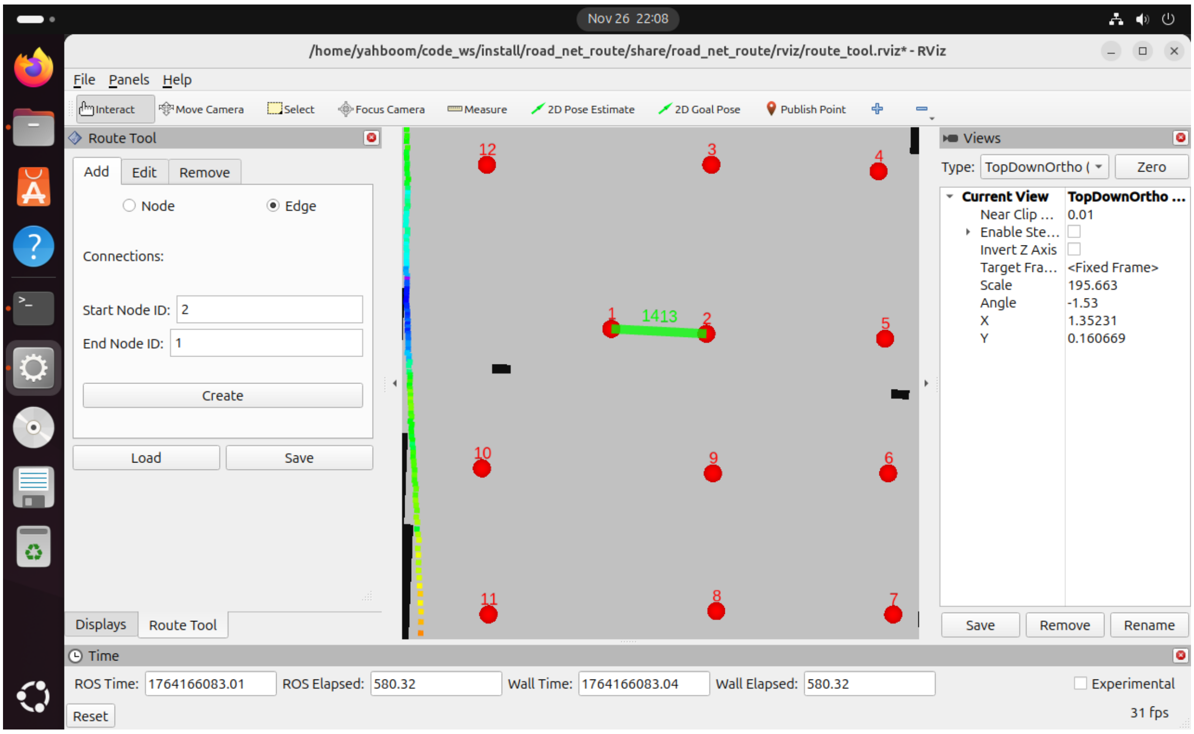
- Select Edge in the `Route Tool` tab.



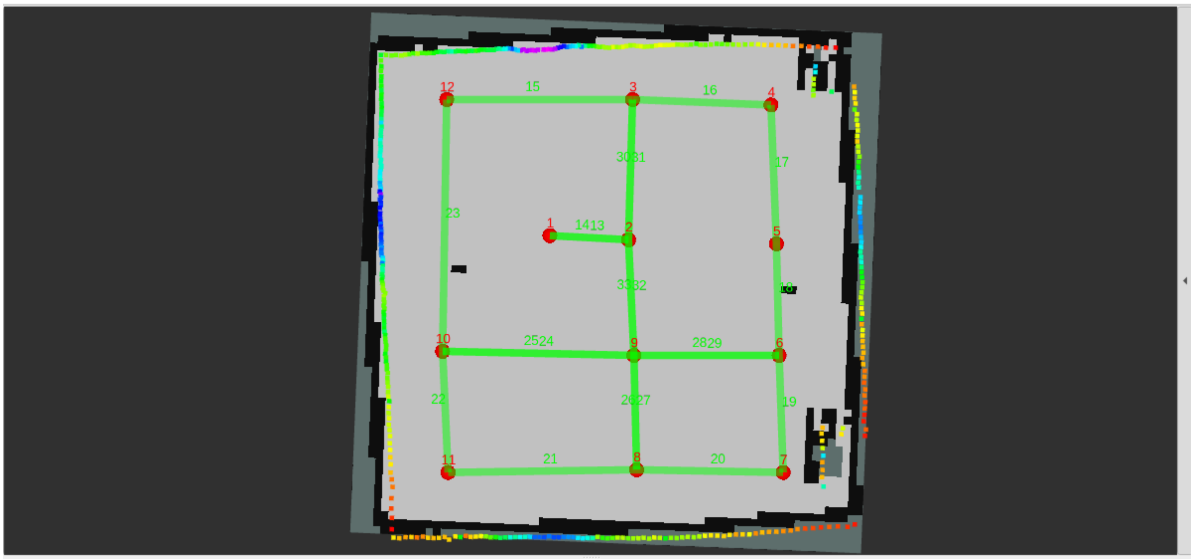
- Enter the starting waypoint in `Start Node ID` and the ending waypoint in `End Node ID`, then click `Create` to create an edge between waypoints.

> [!IMPORTANT]
> **Note!!**
> A road network is a topological structure composed of points and edges. Edges have directions.** Here, the edge created from 1 to 2 is unidirectional. If a bidirectional edge is needed, the edge from 2 to 1 also needs to be created.

- Next, we create the edge from 2 to 1 so that bidirectional travel is possible between road points 1 and 2.

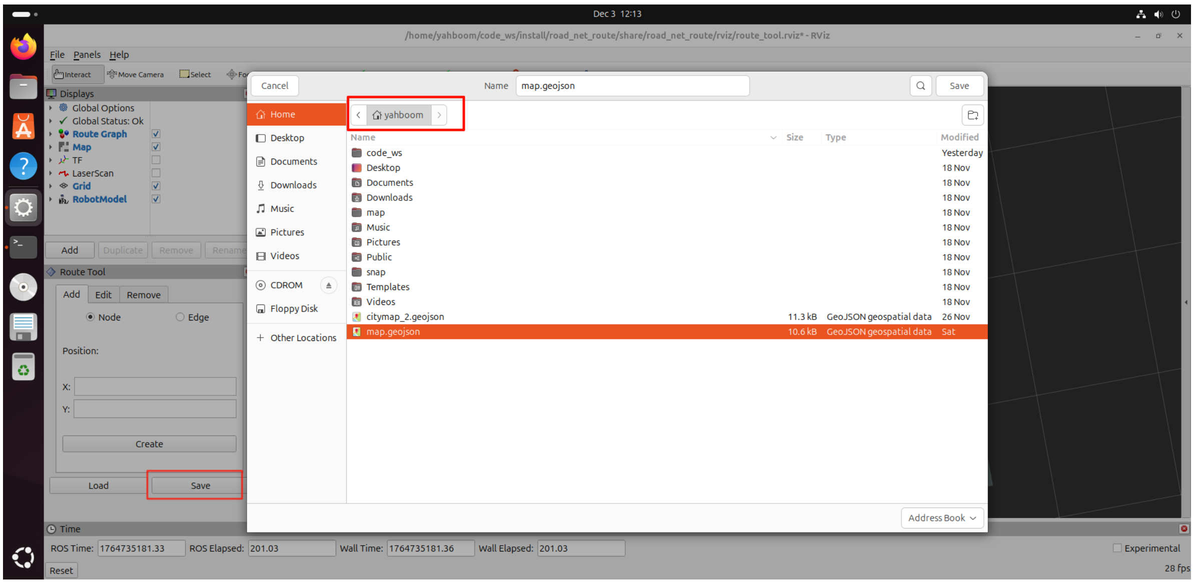


- Repeat the above steps to complete the creation of all edges (whether to create unidirectional or bidirectional edges depends on the user's actual needs).

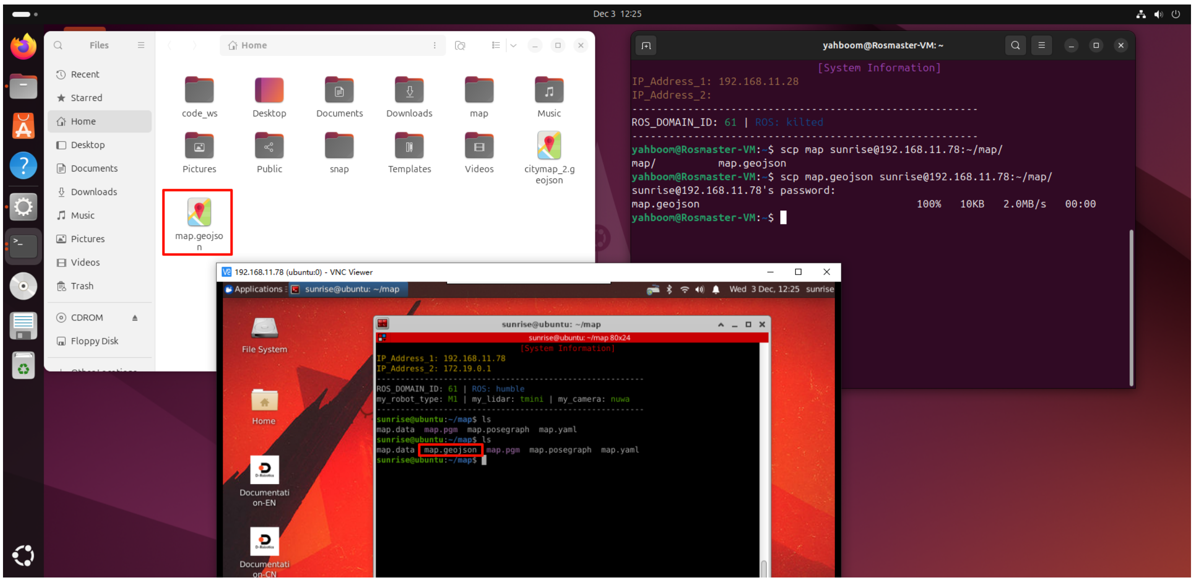


3. Save the road network description file

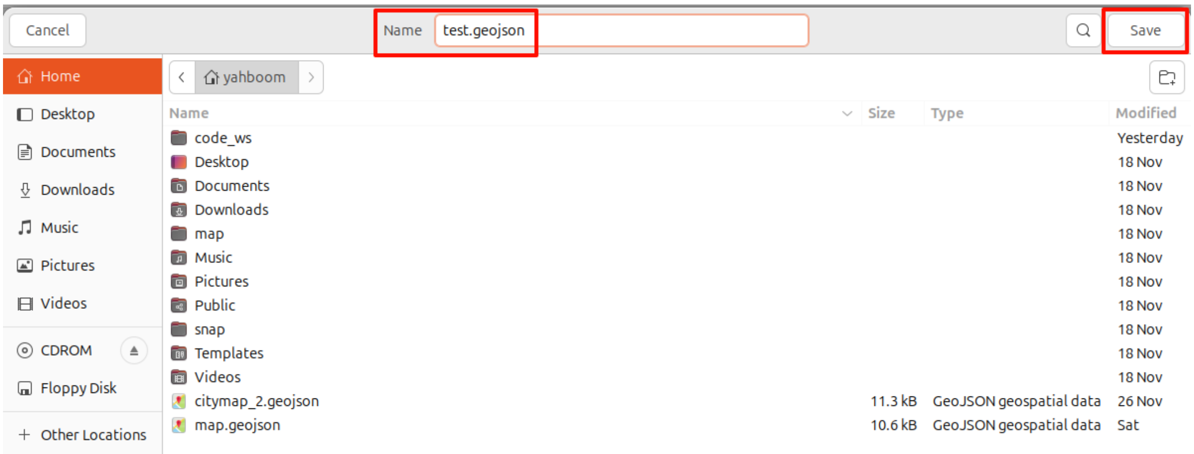
- In `Route In the Tool tab, click Save. In the pop-up window, select the path to save the file. Here, we choose to directly overwrite`map.geojson` and save it in this path.



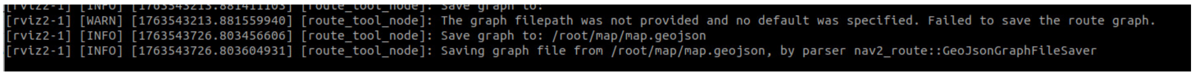
- Transfer it directly to the host machine's `~/map` path using the command `scp map sunrise@192.168.11.78:~/map/` (enter your actual username and IP address). This path is bound to the `/root/map` path inside the Docker container. You can also drag it out from the virtual machine and then transfer it to the host machine.



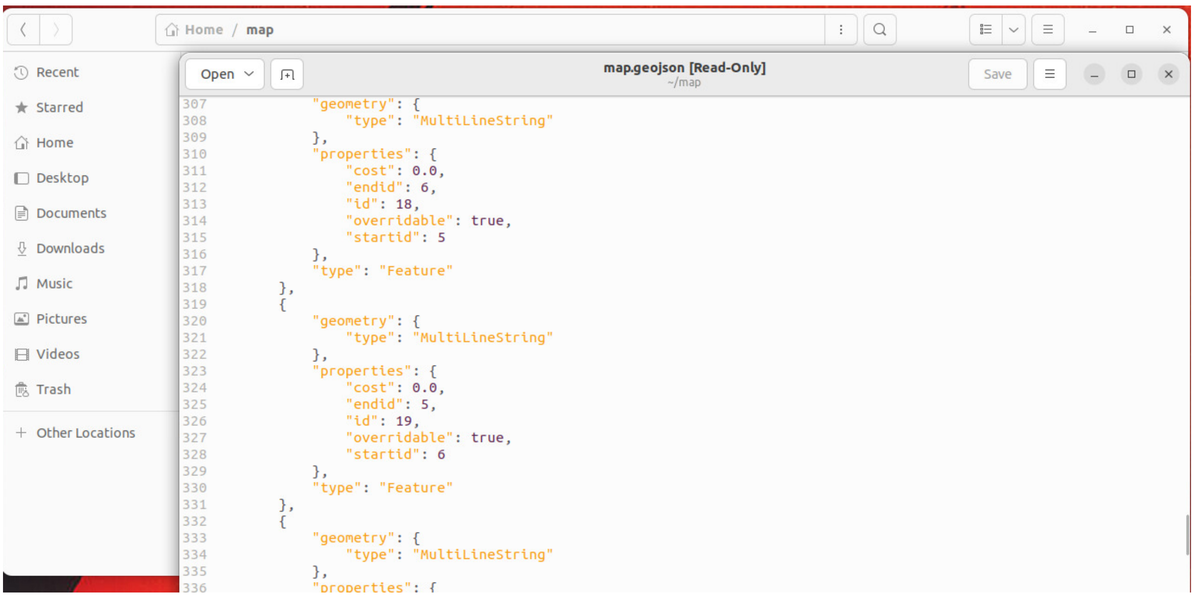
- If you don't choose to overwrite, you can also... Enter the file name in the name field. It is recommended to keep it consistent with the raster map name for easy differentiation. The file extension must be .geojson. Finally, click `save` to save the road network description file.



- After successful saving, the terminal will print a prompt message.



- Afterwards, we can view the `map.geojson` road network description file, which contains a JSON format description of waypoints and edge information.



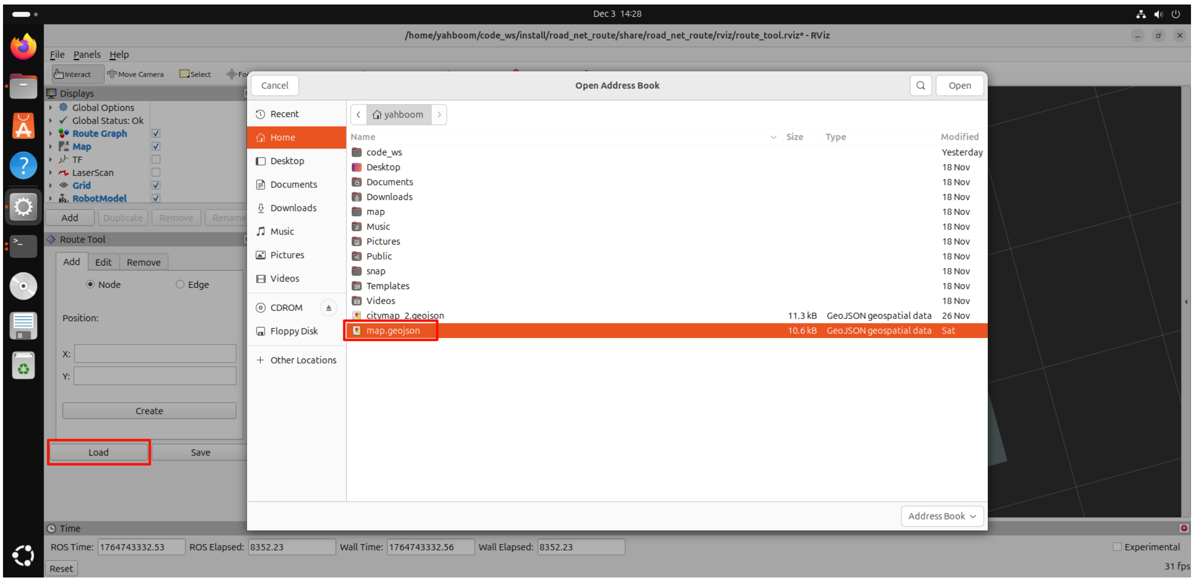
4. Loading and Modifying Road Network Description

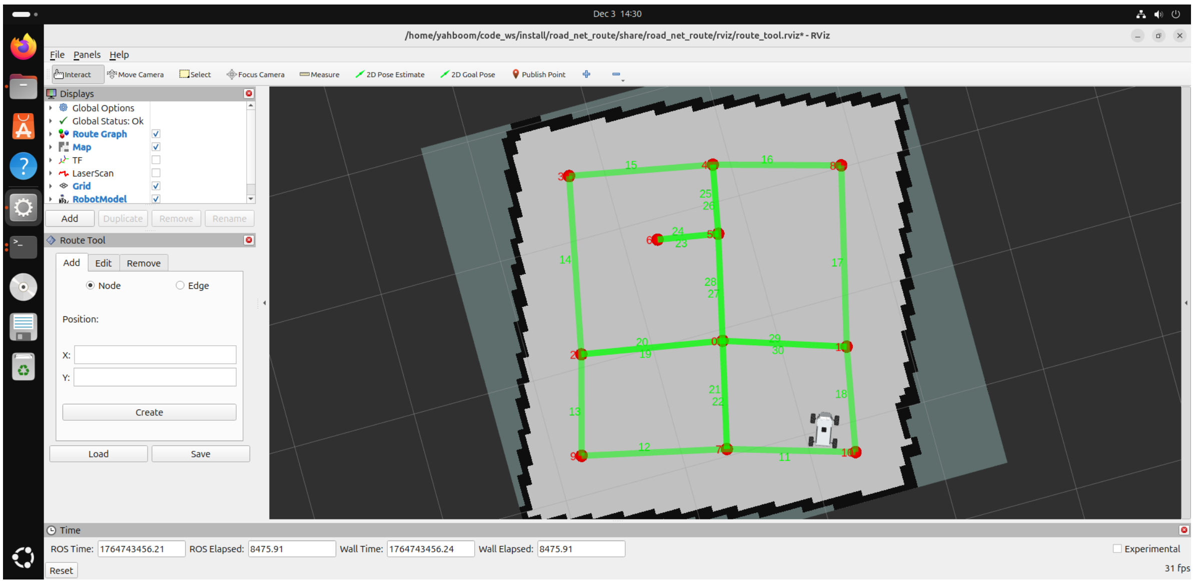
4.1 Loading the Road Network Description File

> [!TIP]
>
> - If you need to modify the previous road network description file later, such as adding, removing, or editing points and edges, you can load the previous road network description file for editing.

Still following step 2.1 after starting the navigation node,

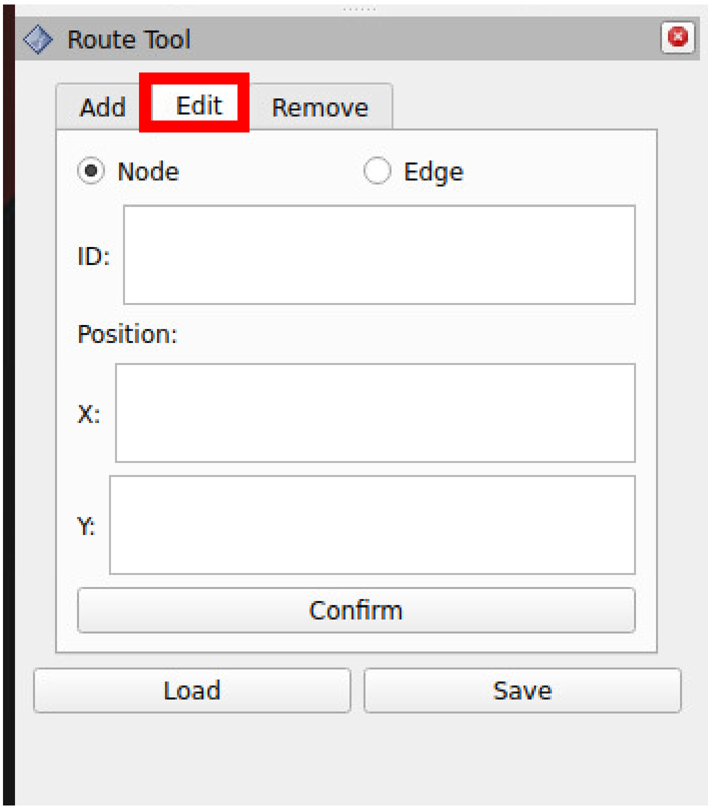
- [x] In the `Route Tool` tab, click Load, then select the road network description file to be edited, and click `open` to load it.





4.2 Editing the Road Network

- If you need to add new waypoints and edges, the operation is the same as before.
- Click `Edit` to edit existing waypoints or edges.



- Click `Remove` to edit existing waypoints or edges.